Modeling DNA methylation profiles through a dynamic equilibrium between methylation and demethylation

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Background

DNA methylation is a heritable epigenetic mark, consisting of the covalent binding of a methyl group to the 5th carbon of cytosines.

DNA methyltransferases (DNMTs) catalyze the transfer of the methyl group from S-adenosyl-l-methionine to cytosine.

Ten-to-eleven translocation dioxygenases (TETs) actively erase 5-methyl cytosines through stepwise oxidative reactions.

Recent studies suggest that methylating and demethylating enzymes cooperate in shaping the DNA methylome, eventually producing heterogeneous methylation profiles among cells [1,2]

Aims of the study

1) To investigate in deep **cell-to-cell heterogeneity** in DNA methylation profiles

2) To model the experimentally observed profiles by assuming the **contemporary presence of DNA methylation and demethylation** at the same DNA region

Methodology

By performing **high-depth amplicon bisulfite sequencing,** we analyzed DNA methylation at the Transcription Start Site surrounding region of the **mouse D-Aspartate Oxidase** (mDDO) gene and of the **human D-Amino Acid Oxidase** (hDAO) gene.

- mDDO DNA methylation was analyzed in three tissues and at different developmental stages.
- hDAO DNA methylation was analyzed in three different brain areas.

In total, we analyzed DNA methylation in **51 samples** related to **17 different conditions.**

For each sample, we grouped the epialleles in methylation classes (MCs), based on the number of methylated cytosines they bore.

Results

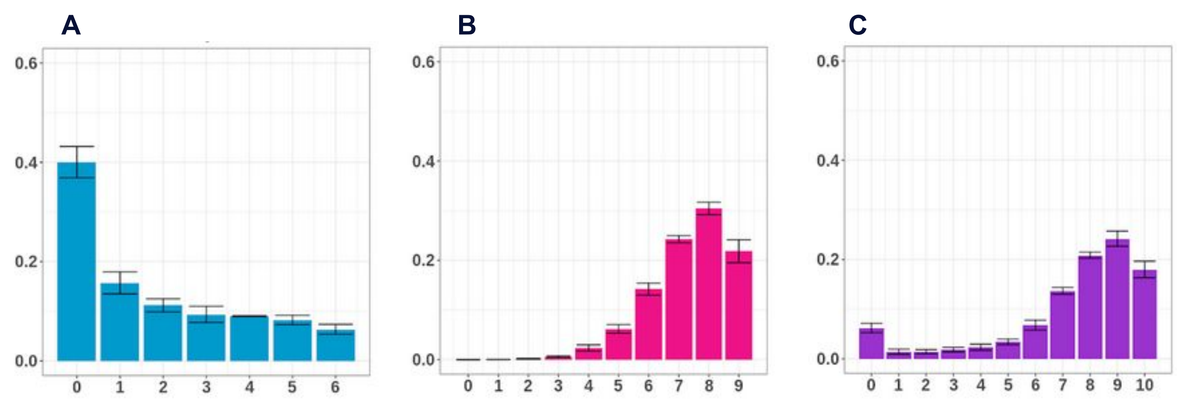

Figure 1: Example of MC profiles of three analyzed condition
For each condition, the MC average frequencies and the standard errors are shown.
It can be noticed that the low methylated classes also have non-zero frequencies in the highly methylated condition and vice-versa (A and B). Notably, some of the analyzed conditions exhibited two prevalent MCs, one being poor and the other being high-methylated (C).

When analyzing the average MC profiles, we noted that in all the conditions almost all the possible MCs were represented, although with different frequencies. This indicated that the DNA methylation status at the analyzed loci **differed among cells**.

However, these cell-to-cell differences were **maintained between different individuals**, which indeed showed very similar MC profiles.

A mathematical model was developed to reproduce the experimentally determined MC profiles.

$$C_n (N-n) P_n = C_{n+1}(n+1)Q_{n+1} \quad (1)$$

C_n and C_{n+1}= number of cell with n and n+1 methylated cytosines

N= total number of cytosines in the considered region

P_n and Q_n =per site rate of DNA methylation and DNA demethylation

Based on the observed pattern of cell-to-cell heterogeneity, MC profiles were modeled as the result of a **dynamic balance between DNA methylation and demethylation.**

Based on the low inter-individual variability, the two processes were supposed to reach a **statistical equilibrium** in the analyzed condition.

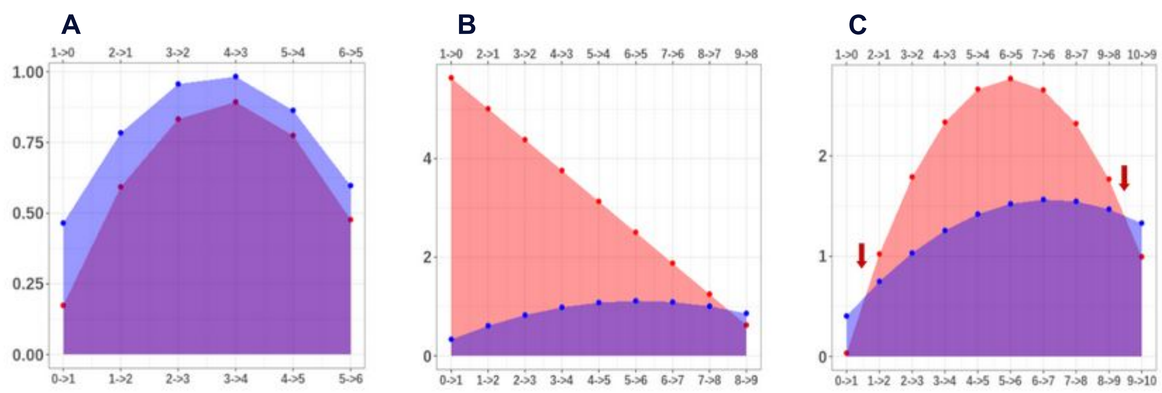

Figure 2: methylation/demethylation rate comparison
For the conditions indicated in Figure1, the corresponding per MC methylation (red) and demethylation (purple) rates are shown. For unimodally distributed profiles, the probability rate of one of the processes (DNA demethylation for A and DNA methylation for B) dominated the other one for all MCs. For Condition 1, the probability rates of the two processes equilibrated at the two MCs (indicated by the red arrows) which exhibited the highest average frequency.

When comparing the experimentally observed MC profiles with their theoretical predictions, we generally observed a **very satisfactory agreement**, as supported by the reduced-$\chi 2$ values well inside of 1-σ statistical uncertainty.

Moreover, the rates of methylation and demethylation per MC, calculated from equation 1, reflected the observed MC profiles, even when they were bimodally distributed.

We modeled DNA methylation and demethylation rates in terms of two contributions:

1: An own methylation or demethylation probability rate (p and q), independent of the configuration of the rest of the region, and averaged on all CpG sites

2: A cooperative methylation and cooperative demethylation probability (α and β), linearly dependent on the total number of methylated and unmethylated sites in the DNA molecule, respectively.

$$P_n = p + \alpha n \quad (2)$$

$$Q_n = q + \beta(N-n) \quad (3)$$

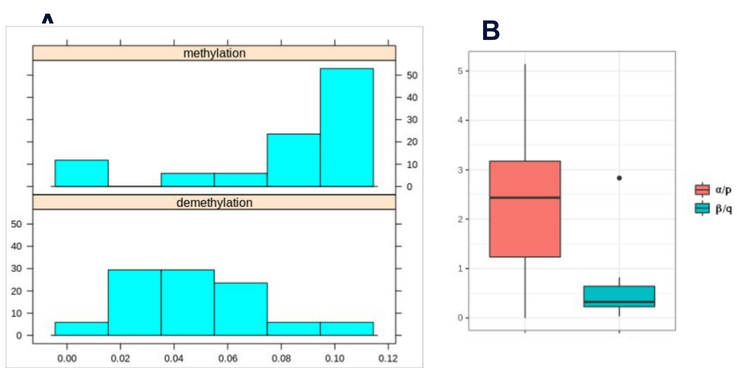

Figure 3: cooperative effect in DNA methylation and demethylation
A: Values of the DNA methylation and demethylation cooperative term in the analyzed conditions. It can be noticed that the cooperative term of DNA methylation mostly assumes higher values than the one of DNA demethylation.
B: Ratio between the cooperative and the independent terms' values for DNA methylation (α/p) and demethylation (β/q). Higher ratios were observed for DNA methylation compared with DNA demethylation. This suggests a higher dependence of the methylation probability rate on the overall methylation of a molecule than for DNA demethylation.

Since in all the analyzed conditions, the parameters' central values were not compatible with a zero value at 2-σ level, meaning that **all of the parameters were statistically necessary** for the description.

The model parameters also suggested that the **cooperative effect of DNA methylation** was in almost all the cases the main contributor to the methylation probability rate of cytosines, whereas the cooperative term of DNA demethylation was never dominant.

Conclusions

The good agreement between the experimental observations and our model theoretical predictions suggest that DNA methylation and demethylation can co-occur in the same genomic region. Furthermore, our model parameters suggest that the rate of both processes is influenced by the methylation status of the surrounding region.

References

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